EE 121B: Principles of Semiconductor Device Design

Lecture 5:

Semiconductor Fundamentals (continued)

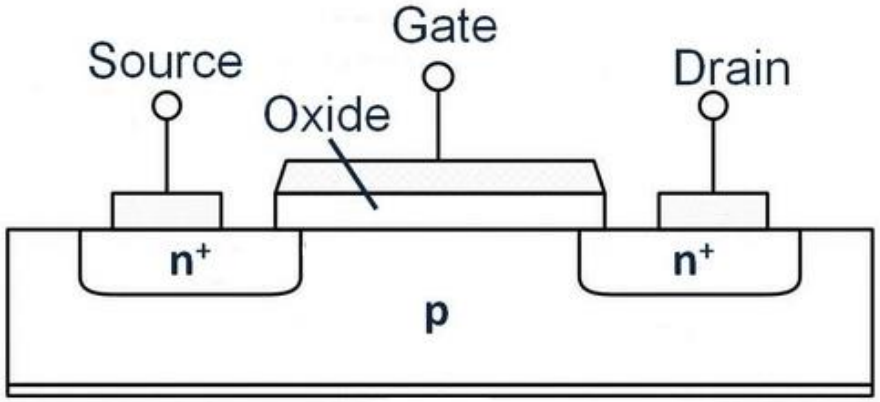
Metal-Semiconductor Contacts

Sam Emaminejad

Course Overview

1. Semiconductor Properties
2. P-N Junction
3. Metal-Semiconductor Contacts
4. MOS Devices
5. Bipolar Junction Transistor

Metal-Oxide-Semiconductor (MOS)
Field-Effect Transistor (FET)



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Lecture 5

OUTLINE

- Semiconductor Fundamentals (cont'd)
  - Continuity equations
  - Minority carrier diffusion equations
  - Minority carrier diffusion length
  - Quasi-Fermi levels
  - Poisson's Equation
  - Work Function
- Metal-Semiconductor Contacts

Reading: Pierret 3.4-3.5, 5.1.2; Hu 4.7, 4.1.3

“Low-Level Injection”

- Often the disturbance from equilibrium is small, such that **the majority-carrier concentration is not affected significantly**:
  - For an n-type material:

$$|\Delta n| = |\Delta p| << n_0 \quad \text{so} \quad n \cong n_0$$

  - For a p-type material:

$$|\Delta n| = |\Delta p| << p_0 \quad \text{so} \quad p \cong p_0$$

However, the minority carrier concentration can be significantly affected.

Indirect Recombination Rate

Suppose excess carriers are introduced into an n-type Si sample (*e.g.* by temporarily shining light onto it) at time $t = 0$

$$\left.\frac{\partial p}{\partial t}\right|_{R-G} = -c_p N_T \Delta p \equiv -\frac{\Delta p}{\tau_p}$$

$$\text{where } \tau_p \equiv \frac{1}{c_p N_T}$$

Minority Carrier (Recombination) Lifetime

$$\tau_p \equiv \frac{1}{c_p N_T} \qquad \tau_n \equiv \frac{1}{c_n N_T}$$

The ***minority carrier lifetime*** τ is the average time an excess minority carrier "survives" in a sea of majority carriers

τ ranges from 1 ns to 1 ms in Si and depends on the density of metallic impurities (contaminants) such as Au and Pt, and the density of crystalline defects. These impurities/defects give rise to localized energy states deep within the band gap. Such ***deep traps*** capture electrons or holes to facilitate recombination and are called ***recombination-generation centers.***

Relaxation to Equilibrium State

Consider a semiconductor with no current flow in which thermal equilibrium is disturbed by the sudden creation of excess holes and electrons. The system will relax back to the equilibrium state via the R-G mechanism:

$$\frac{\partial n}{\partial t} = -\frac{\Delta n}{\tau_n}$$

for electrons in p-type material

$$\frac{\partial p}{\partial t} = -\frac{\Delta p}{\tau_p}$$

for holes in n-type material

Example: Photoconductor

Consider a sample of Si doped with 10^{16} cm^{-3} boron, ← Acceptor p-type with recombination lifetime 1 μs. It is exposed continuously to light, such that electron-hole pairs are generated throughout the sample at the rate of 10^{20} per cm^3 per second, *i.e.* the **generation rate G_L** = 10^{20}/cm^3/s

What are p_0 and n_0 ?

$$p_0 = N_A = 10^{16}\text{cm}^{-3}$$
$$n_0 = \frac{n_i^2}{N_A} = \frac{10^{20}}{10^{16}} = 10^4\text{cm}^{-3}$$

What are Δn and Δp ?

$$G_L = \frac{\partial \Delta n}{\partial t} = \frac{\partial \Delta p}{\partial t} = \frac{\Delta n}{\tau_n}$$
$$\Delta n = G_L \tau_n = 10^{20} \times 10^{-6} = 10^{14}\text{cm}^{-3} = \Delta p$$

(Hint: In steady-state, generation rate equals recombination rate.)

What are *p* and *n* ?

$$p = p_0 + \Delta p = 10^{16} + 10^{14} = 10^{16}\text{cm}^{-3}$$
$$n = n_0 + \Delta n = 10^{4} + 10^{14} = 10^{14}\text{cm}^{-3}$$

What is the *np* product ?

$$np = 10^{16} \times 10^{14} = 10^{30}\text{cm}^{-6} \gg n_i^2 \ (= 10^{20}\text{cm}^{-6} \text{ at 300K})$$

Note: The *np* product can be very different from n_i^2.

Net Recombination Rate (General Case)

For arbitrary injection levels, the net rate of carrier recombination is:

$$-\frac{\partial \Delta n}{\partial t} = -\frac{\partial \Delta p}{\partial t} = \frac{pn - n_i^2}{\tau_p (n + n_1) + \tau_n (p + p_1)}$$

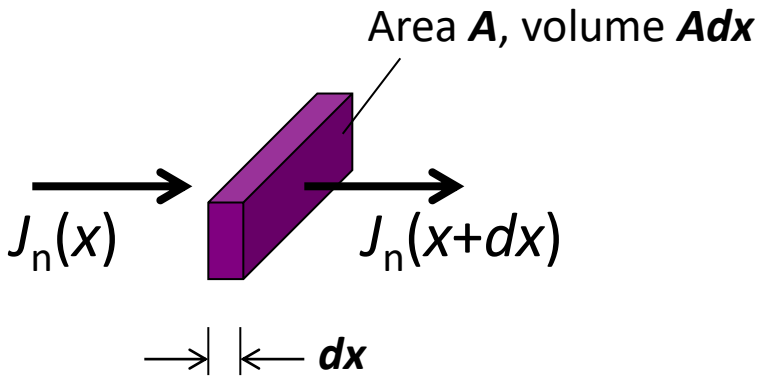
where $n_1 \equiv n_i e^{(E_T - E_i)/kT}$ and $p_1 \equiv n_i e^{(E_i - E_T)/kT}$

If $pn > n_i^2 \longrightarrow \frac{\partial \Delta p}{\partial t} \left(= \frac{\partial \Delta n}{\partial t} \right) < 0 \longrightarrow$ Net Recombination

If $pn < n_i^2 \longrightarrow \frac{\partial \Delta p}{\partial t} \left(= \frac{\partial \Delta n}{\partial t} \right) > 0 \longrightarrow$ Net Generation

Derivation of Continuity Equation

- Consider carrier-flux into/out-of an infinitesimal volume:



Change in # of electrons in the volume

Difference between # electrons entering and exiting the volume

$$Adx\left(\frac{\partial n}{\partial t}\right) = -\frac{1}{q}\left[J_n(x)A - J_n(x+dx)A\right] - \frac{\Delta n}{\tau_n}Adx$$

of electrons annihilate within volume

$$J_n(x+dx) = J_n(x) + \frac{\partial J_n(x)}{\partial x} dx$$

$$\Rightarrow \quad \frac{\partial n}{\partial t} = \frac{1}{q} \frac{\partial J_n(x)}{\partial x} - \frac{\Delta n}{\tau_n}$$

Rate at which electrons and holes are generated in pairs due to irradiation

Continuity Equations:

$$\frac{\partial n}{\partial t} = \frac{1}{q} \frac{\partial J_n(x)}{\partial x} - \frac{\Delta n}{\tau_n} + G_L$$

$$\frac{\partial p}{\partial t} = -\frac{1}{q} \frac{\partial J_p(x)}{\partial x} - \frac{\Delta p}{\tau_p} + G_L$$

Derivation of Minority Carrier Diffusion Equation

- **The minority carrier diffusion equations are derived from the general continuity equations, and are applicable only for minority carriers.**
- **Simplifying assumptions:**

1. The electric field is small, such that

$$J_n = q\mu_n n\mathcal{E} + qD_n \frac{\partial n}{\partial x} \cong qD_n \frac{\partial n}{\partial x}$$ (Definition) in p-type material

$$J_p = q\mu_p p\mathcal{E} - qD_p \frac{\partial p}{\partial x} \cong -qD_p \frac{\partial p}{\partial x}$$ in n-type material

2. *Majority carrier concentration is* independent of *x*

→ *i.e.* uniform doping

3. Low-level injection conditions prevail

Recall that $n = n_0 + \Delta n$

- Starting with the continuity equation for electrons: (p-type material)

$$\frac{\partial n}{\partial t} = \frac{1}{q}\frac{\partial J_n(x)}{\partial x} - \frac{\Delta n}{\tau_n} + G_L$$

$$\frac{\partial (n_0 + \Delta n)}{\partial t} = \frac{1}{q}\frac{\partial}{\partial x}\left[qD_n \frac{\partial (n_0 + \Delta n)}{\partial x} \right] - \frac{\Delta n}{\tau_n} + G_L$$

$$\frac{\partial \Delta n}{\partial t} = D_n \frac{\partial^2 \Delta n}{\partial x^2} - \frac{\Delta n}{\tau_n} + G_L$$

Carrier Concentration Notation

- **The subscript "n" or "p" is used to explicitly denote n-type or p-type material, *e.g.***

  p_n is the hole (minority-carrier) concentration in n-type mat'l

  n_p is the electron (minority-carrier) concentration in p-type mat'l

- **Thus the minority carrier diffusion equations are**

$$\frac{\partial \Delta n_p}{\partial t} = D_n \frac{\partial^2 \Delta n_p}{\partial x^2} - \frac{\Delta n_p}{\tau_n} + G_L$$

$$\frac{\partial \Delta p_n}{\partial t} = D_p \frac{\partial^2 \Delta p_n}{\partial x^2} - \frac{\Delta p_n}{\tau_p} + G_L$$

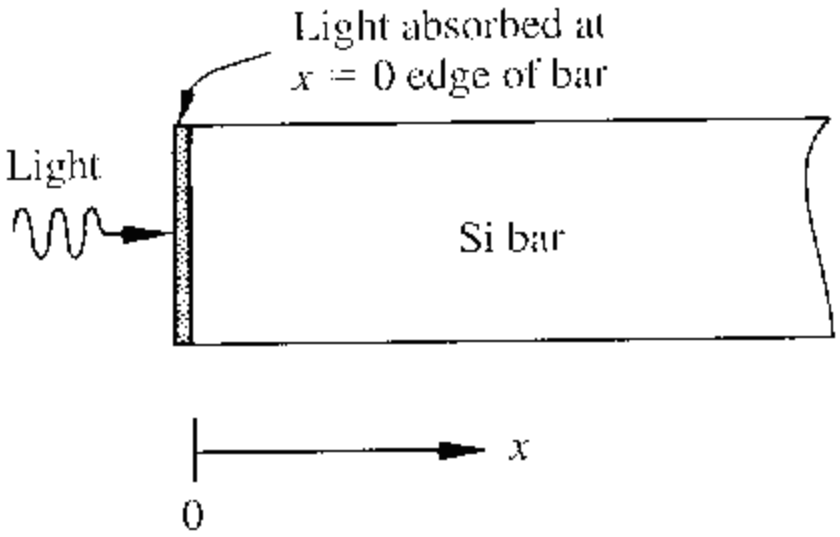
Simplifications (Special Cases)

- **Steady state:** $\frac{\partial \Delta n_p}{\partial t} = 0 \qquad \frac{\partial \Delta p_n}{\partial t} = 0$
- **No diffusion current:** $D_n \frac{\partial^2 \Delta n_p}{\partial x^2} = 0 \qquad D_p \frac{\partial^2 \Delta p_n}{\partial x^2} = 0$
- **No R-G:** $\frac{\Delta n_p}{\tau_n} = 0 \qquad \frac{\Delta p_n}{\tau_p} = 0$
- **No light:** $G_L = 0$

Example

$$\frac{\partial \Delta p_n}{\partial t} = D_p \frac{\partial^2 \Delta p_n}{\partial x^2} - \frac{\Delta p_n}{\tau_p} + G_L$$

- **Consider an n-type Si sample illuminated at one end:**
  - constant minority-carrier injection at $x = 0$ $\Delta p_n(0) = \Delta p_{n0}$
  - steady state; no light absorption for $x > 0$ $G_L = 0$



$$0 = \frac{\partial \Delta p_n}{\partial t} = D_p \frac{\partial^2 \Delta p_n}{\partial x^2} - \frac{\Delta p_n}{\tau_p} = 0$$

$$\frac{\partial^2 \Delta p_n}{\partial x^2} = \frac{\Delta p_n}{\tau_p D_p} = \frac{\Delta p_n}{L_p^2}$$

L_p is the ***hole diffusion length***: $L_p \equiv \sqrt{D_p \tau_p}$

The general solution to the equation $\frac{\partial^2 \Delta p_n}{\partial x^2} = \frac{\Delta p_n}{L_p^{\,2}}$

is $\Delta p_n(x) = Ae^{-x/L_p} + Be^{x/L_p}$

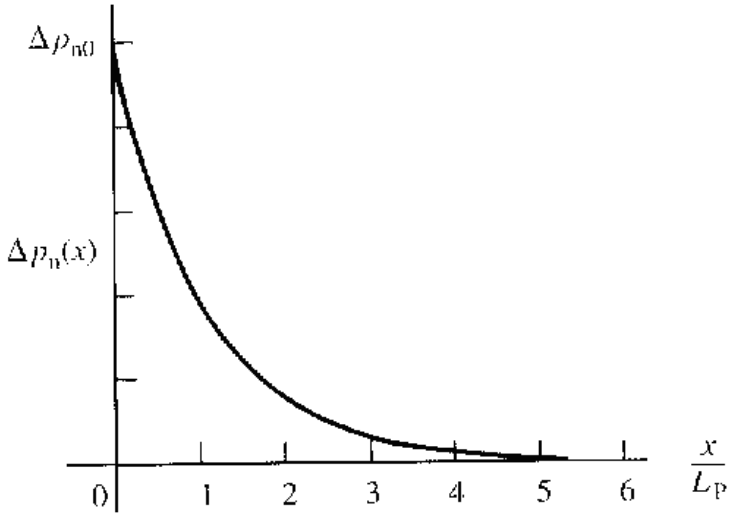
where *A, B* are constants determined by boundary conditions:

$\Delta p_n(\infty) = 0 \longrightarrow$ B = 0

$\Delta p_n(0) = \Delta p_{n0}$ = A

Therefore, the solution is

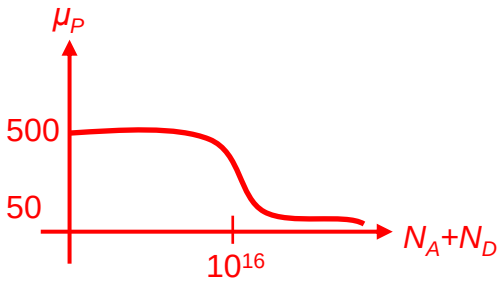
$$\Delta p_n(x) = \Delta p_{n0} e^{-x/L_p}$$



Minority Carrier Diffusion Length

- Physically, L_p and L_n represent the average distance that minority carriers can diffuse into a sea of majority carriers before being annihilated.

- Example: $N_D = 10^{16}$ cm^{-3}; $\tau_p = 10^{-6}$ s



$$L_P = \sqrt{D_P \tau_P}$$

For $N_D + N_A = 10^{16}\text{cm}^{-3}$, $\mu_P = 450\ \text{cm/(V}\cdot\text{s)}$

$$D_P = \frac{kT}{q}\mu_P = 0.026(450) = 12\ \text{cm}^2/\text{s}$$

$$L_P = \sqrt{12 \times 10^{-6}} = 3.5 \times 10^{-3}\text{cm} = 35\mu\text{m}$$

Summary: Continuity Equations

- The ***continuity equations*** are established based on conservation of carriers, and therefore hold generally:

$$\frac{\partial n}{\partial t} = \frac{1}{q}\frac{\partial J_n(x)}{\partial x} - \frac{\Delta n}{\tau_n} + G_L \qquad \frac{\partial p}{\partial t} = -\frac{1}{q}\frac{\partial J_n(x)}{\partial x} - \frac{\Delta p}{\tau_p} + G_L$$

- The ***minority carrier diffusion equations*** are derived from the continuity equations, specifically for minority carriers under certain conditions (small *E*-field, low-level injection, uniform doping profile):

$$\frac{\partial \Delta n_p}{\partial t} = D_N \frac{\partial^2 \Delta n_p}{\partial x^2} - \frac{\Delta n_p}{\tau_n} + G_L \qquad \frac{\partial \Delta p_n}{\partial t} = D_P \frac{\partial^2 \Delta p_n}{\partial x^2} - \frac{\Delta p_n}{\tau_p} + G_L$$

Quasi-Fermi Levels

- Whenever $\Delta n = \Delta p \neq 0$, $np \neq n_i^2$. However, we would like to preserve and use the relations:

$$n = n_i e^{(E_F - E_i)/kT} \qquad p = n_i e^{(E_i - E_F)/kT}$$

- These equations imply $np = n_i^2$, however. The solution is to introduce two ***quasi-Fermi levels*** F_N **and** F_P such that

$$n = n_i e^{(F_N - E_i)/kT} \qquad p = n_i e^{(E_i - F_P)/kT}$$

Example: Quasi-Fermi Levels

Consider a Si sample with $N_D = 10^{17}\,\text{cm}^{-3}$ and $\Delta n = \Delta p = 10^{14}\,\text{cm}^{-3}$.

What are p and n ?

$$p_0 = 10^3\text{cm}^{-3}, \qquad p = p_0 + \Delta p = 10^3 + 10^{14} = 10^{14}\text{cm}^{-3}$$
$$n_0 = 10^{17}\text{cm}^{-3}, \qquad n = n_0 + \Delta n = 10^{17} + 10^{14} = 10^{17}\text{cm}^{-3}$$

What is the np product ?

$$10^{17}10^{14} = 10^{31}\text{cm}^{-6} \gg n_i^2$$

$$n = n_i e^{(F_N - E_i)/kT}$$

$$p = n_i e^{(E_i - F_P)/kT}$$

$n_i = 10^{10}\text{cm}^{-3}$ in Si at 300K

- ***Find F_N and F_P :***

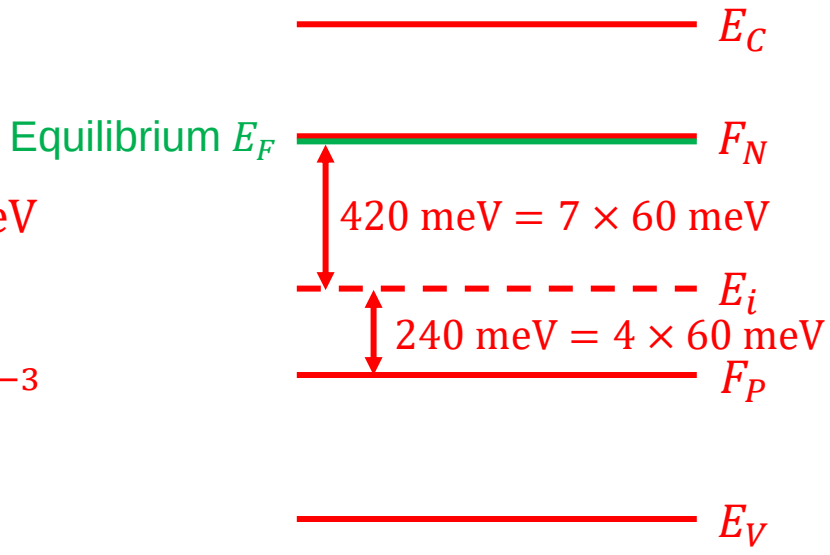
$$F_N \equiv E_i + kT \ln\left(\frac{n}{n_i}\right)$$

10^{17}cm^{-3}

$7kT\ln(10) = 7 \times 0.06\ \text{eV}$

$$F_P \equiv E_i - kT \ln\left(\frac{p}{n_i}\right)$$

10^{14}cm^{-3}



Gauss' Law

$$\frac{d\mathcal{E}}{dx} = \frac{\rho}{\varepsilon_s}$$

$$\mathcal{E} = -\frac{dV}{dx}$$

$$-\frac{d^2V}{dx^2} = \frac{\rho}{\varepsilon_s}$$

$\varepsilon_0 = 8.85 \times 10^{-14} (\mathrm{F}/\mathrm{cm})$

$\varepsilon_{r,si} \approx 12$

$\varepsilon_{r,SiO_2} \approx 4$

ε_s : permittivity (F/cm)

ρ : charge density (C/cm^3)

Charge Density in a Semiconductor

- Assuming the dopants are completely ionized:

$$\rho = q\,(p - n + ND - NA)$$

Charge density — Mobile — Immobile

Example: Suppose a semiconductor is doped with phosphorus (10^{15} cm^{-3}), but electrons diffuse away (due to diffusion, for example), what is ρ?

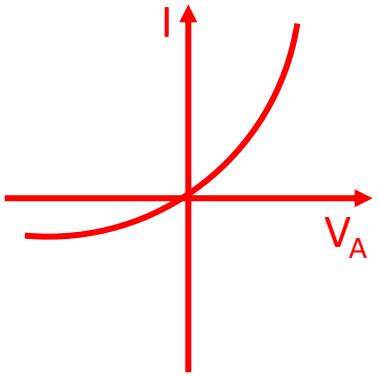
$$\rho = qN_D = (1.6 \times 10^{-19})(10^{15}) = 1.6 \times 10^{-4}\ \mathrm{C/cm}^{-3}$$

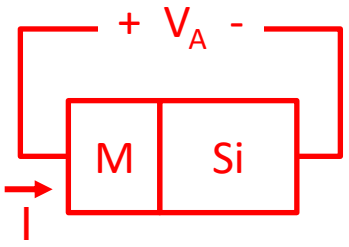
$$\frac{d\mathcal{E}}{dx} = \frac{\rho}{\varepsilon_s}$$

Metal-Semiconductor Contacts

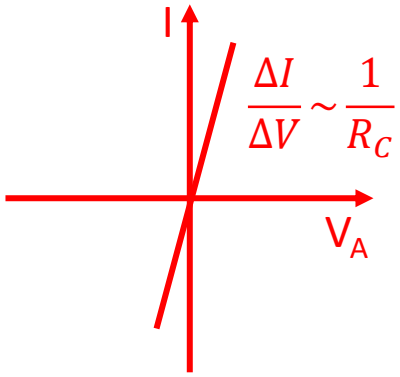
There are 2 kinds of metal-semiconductor contacts:

- rectifying
  "Schottky diode"





- non-rectifying
  "ohmic contact"



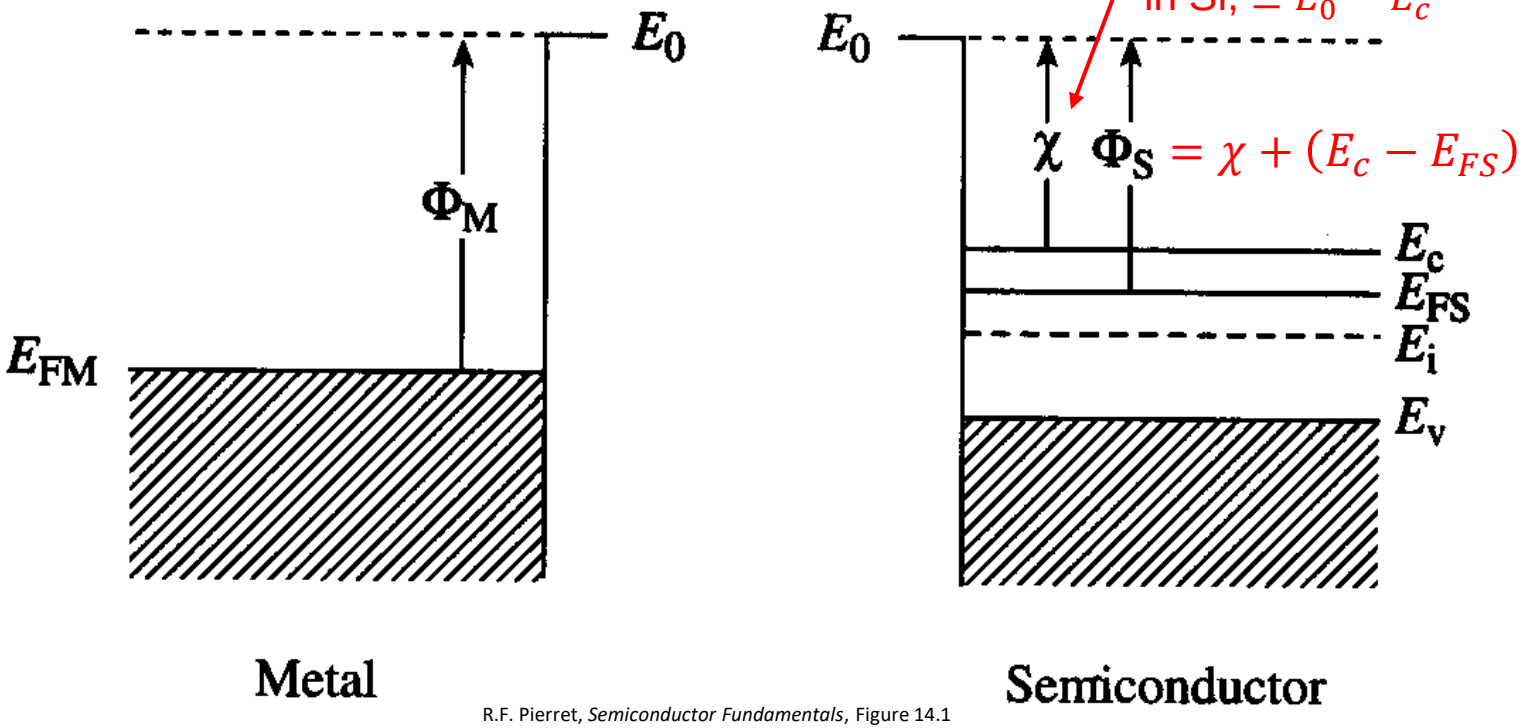
Units: eV, $\Phi = E_0 - E_F$

Work Function

Minimum energy needed to remove an electron from a solid

E_0: vacuum energy level

χ: electron affinity, 4.0 eV in Si, $\equiv E_0 - E_c$



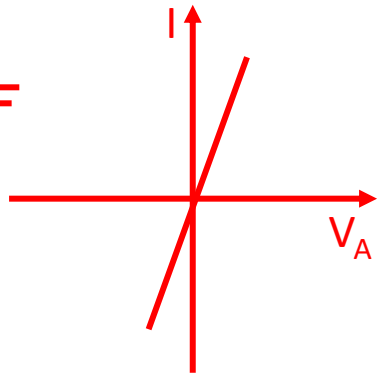
R.F. Pierret, *Semiconductor Fundamentals*, Figure 14.1

Φ_M: metal work function

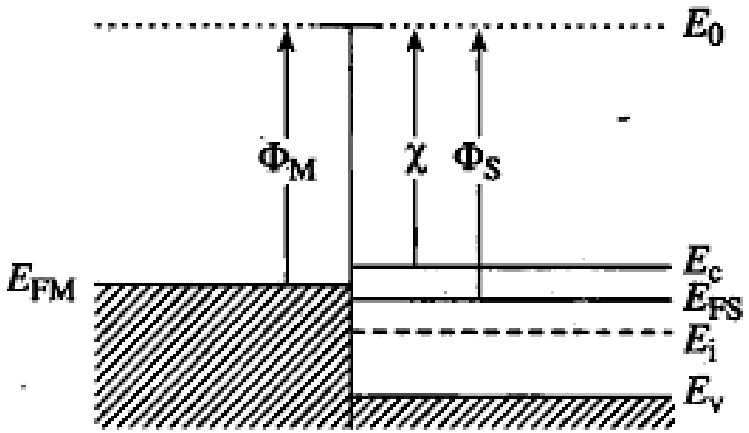
Φ_S: semiconductor work function

Ideal M-S Contact: $\Phi_M < \Phi_S$, n-type



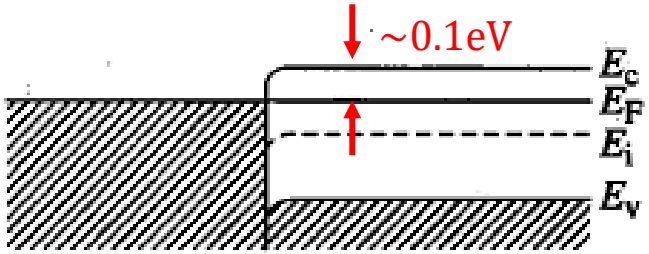


Band diagram instantly after contact formation:



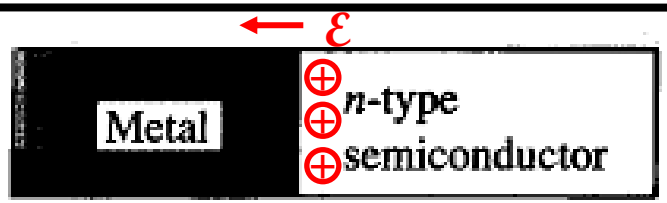
"Ohmic contact"

Equilibrium band diagram:

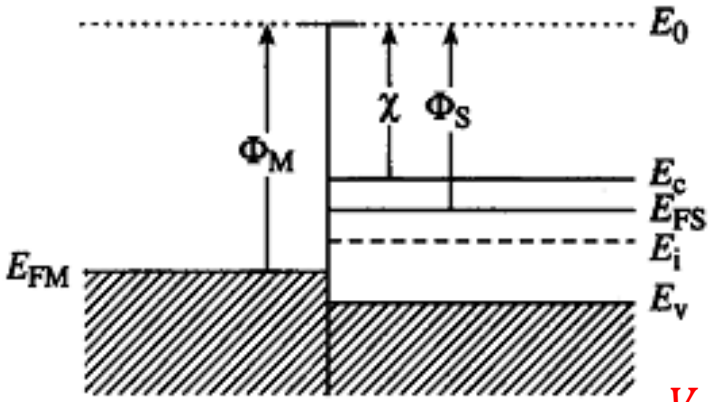


R.F. Pierret, *Semiconductor Fundamentals*, Fig. 14.2

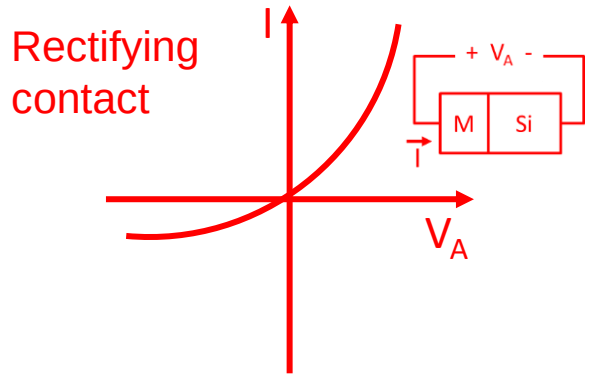
Ideal M-S Contact: $\Phi_M > \Phi_S$, n-type



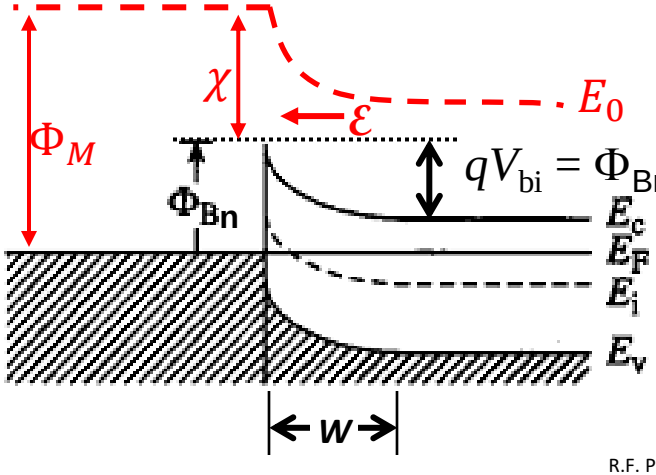
Band diagram instantly after contact formation:



V_{bi} = built-in potential



Equilibrium band diagram:



Flat band

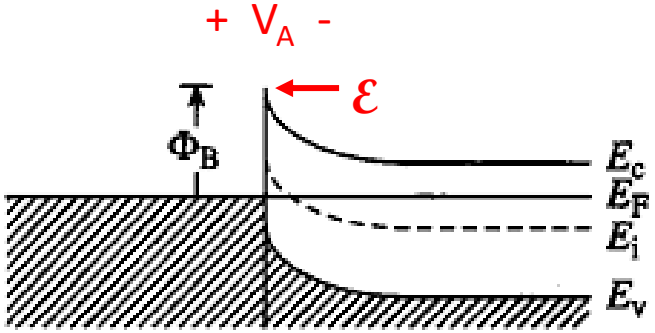
***Schottky Barrier Height*:**

$$\Phi_{Bn} = \Phi_M - \chi$$

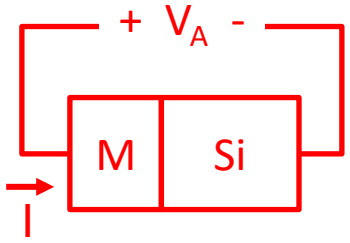
R.F. Pierret, *Semiconductor Fundamentals*, Fig. 14.2

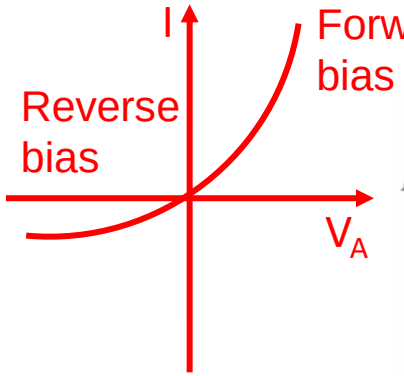
Voltage Drop across the M-S Contact

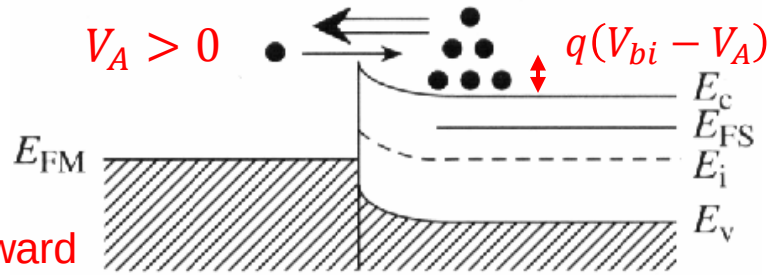
- Under equilibrium conditions ($V_A = 0$), the voltage drop across the semiconductor depletion region is the built-in voltage V_{bi}.

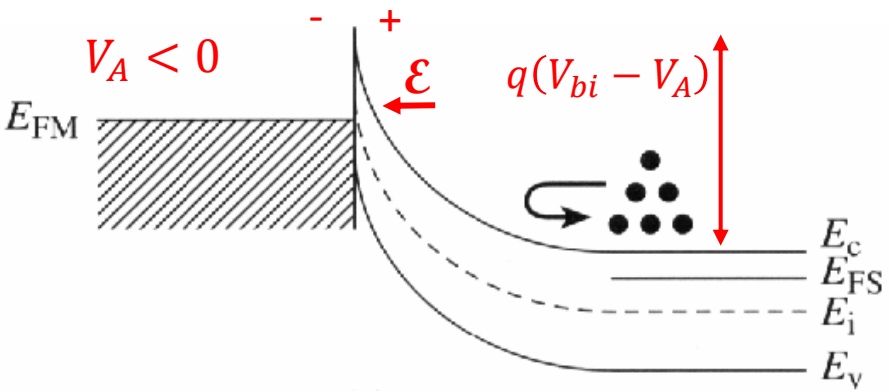


- If $V_A \neq 0$, the voltage drop across the semiconductor depletion region is $V_{bi} - V_A$.





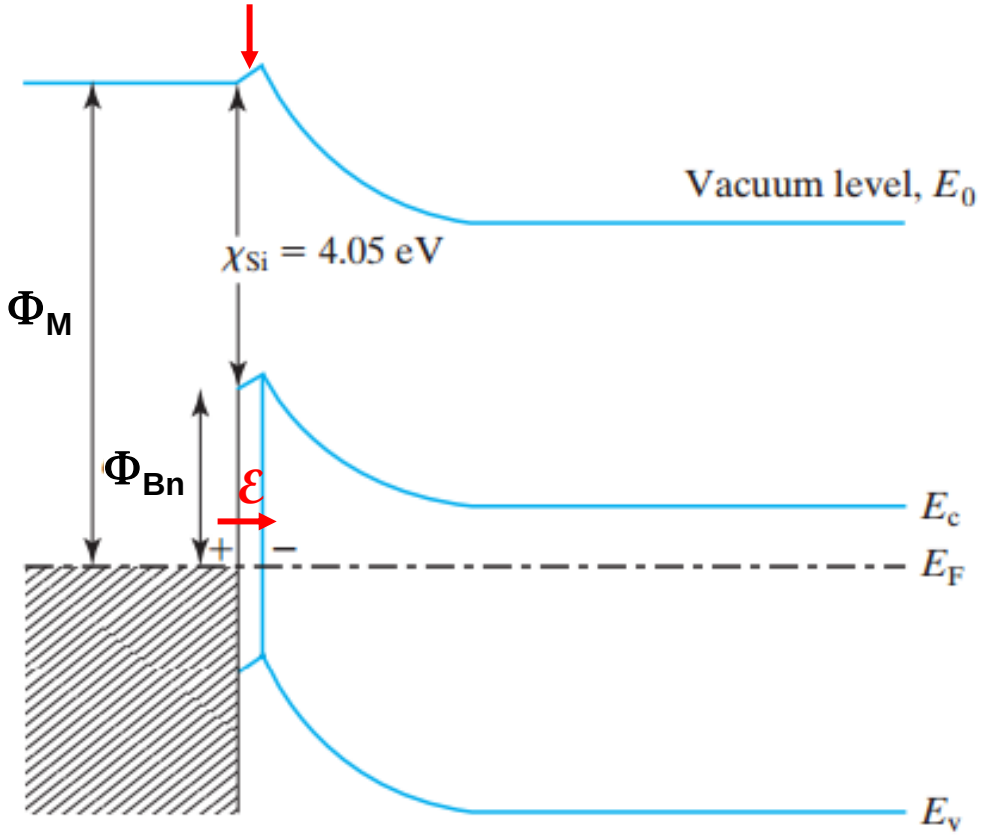




R.F. Pierret, *Semiconductor Fundamentals*, Fig. 14.3

Effect of Interface States on Φ_{Bn}

Band-bending due to dipole induced by electrons trapped at the M-S interface



C. C. Hu, *Modern Semiconductor Devices for ICs*, Figure 4-35

- **Ideal M-S contact:**

$$\Phi_{Bn} = \Phi_M - \chi$$

- **Real M-S contacts:**

A high density of allowed energy states in the band gap at the M-S interface "pins" E_F to be within the range 0.4 eV to 0.9 eV below E_c

Schottky Barrier Heights: Metal on Si

Metal	Er	Ti	Ni	W	Mo	Pt
Φ_M (eV)	3.12	4.3	4.7	4.6	4.6	5.6
Φ_{Bn} (eV)	0.44	0.5	0.61	0.67	0.68	0.73
Φ_{Bp} (eV)	0.68	0.61	0.51	0.45	0.42	0.39

- Φ_{Bn} tends to increase with increasing metal work function

Schottky Barrier Heights: Silicide on Si

Silicide	$ErSi_{1.7}$	$TiSi_2$	$CoSi_2$	NiSi	WSi_2	PtSi
Φ_M (eV)	3.78	4.18	4.6	4.65	4.7	5
Φ_{Bn} (eV)	0.3	0.6	0.64	0.65	0.65	0.84
Φ_{Bp} (eV)	0.8	0.52	0.48	0.47	0.47	0.28

Silicide-Si interfaces are more stable than metal-silicon interfaces and hence are much more prevalent in ICs. After metal is deposited on Si, a thermal annealing step is applied to form a silicide-Si contact. The term ***metal-silicon contact*** includes silicide-Si contacts.

The Depletion Approximation

The semiconductor is depleted of mobile carriers to a depth W

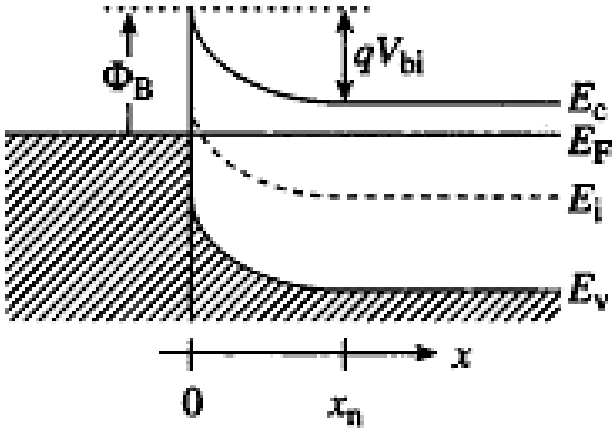
n and p each is much less than the net dopant concentration (N_D-N_A) within the depleted region

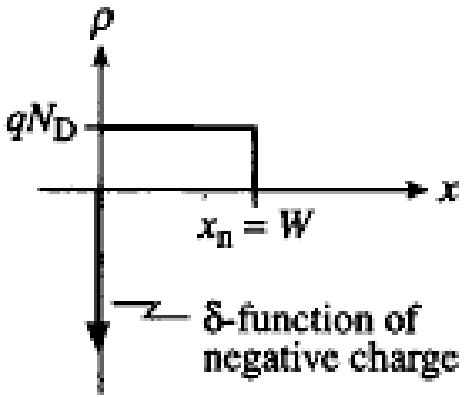
$\Rightarrow$ In the depleted region ($0 \leq x \leq W$):

$$\rho = q\,(N_D - N_A)$$

Beyond the depleted region ($x > W$):

$$\rho = 0$$





R.F. Pierret, *Semiconductor Fundamentals*, Fig. 14.4

Electrostatics

Poisson's equation
$\rho(x) \rightarrow \mathcal{E}(x) \rightarrow V(x)$

- Poisson's equation: $\dfrac{\partial \mathcal{E}}{\partial x} = \dfrac{\rho}{\varepsilon_s} \cong \dfrac{qN_D}{\varepsilon_s}$

$$\mathcal{E}(x) = \int \frac{\rho}{\varepsilon_s} dx = Ax + B$$

$$\text{slope} = A = \frac{d\mathcal{E}}{dx} = \frac{qN_D}{\varepsilon_s}$$

Boundary condition:

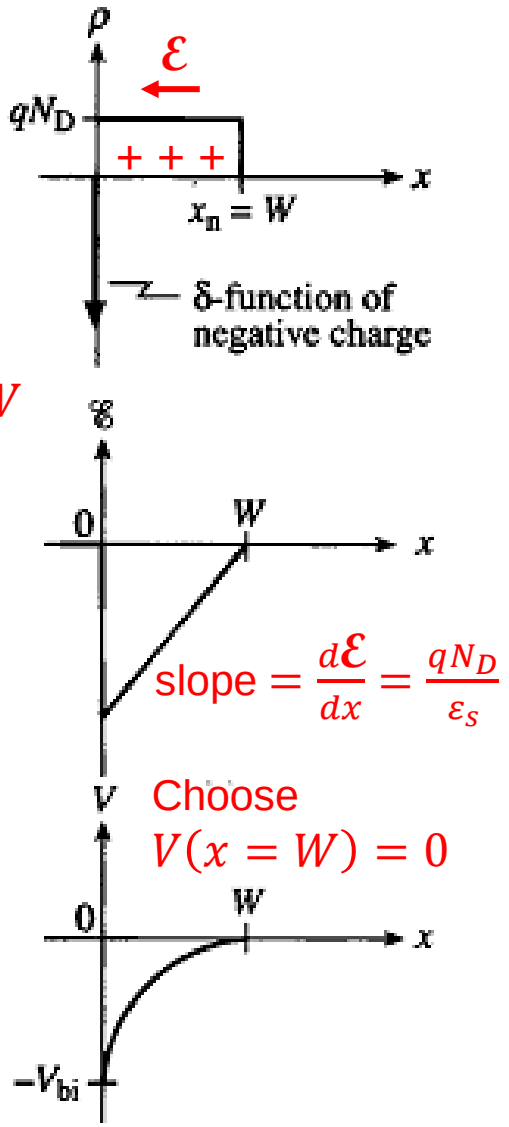
$$\mathcal{E}(W) = 0 \rightarrow \mathcal{E}(W) = \frac{qN_D}{\varepsilon_s} W + B = 0 \rightarrow B = -\frac{qN_D}{\varepsilon_s} W$$

- The solution is: $\mathcal{E}(x) = -\dfrac{qN_D}{\varepsilon_s}(W - x)$

$$\mathcal{E} = -\frac{dV}{dx}$$

$$V(x) = -\int \mathcal{E}(x')dx' = \int \frac{qN_D}{\varepsilon_s}(W - x)dx$$

$$= -\frac{qN_D}{2\varepsilon_s}(W - x)^2$$



R.F. Pierret, *Semiconductor Fundamentals*, Fig. 14.4

Depletion Width, *W*

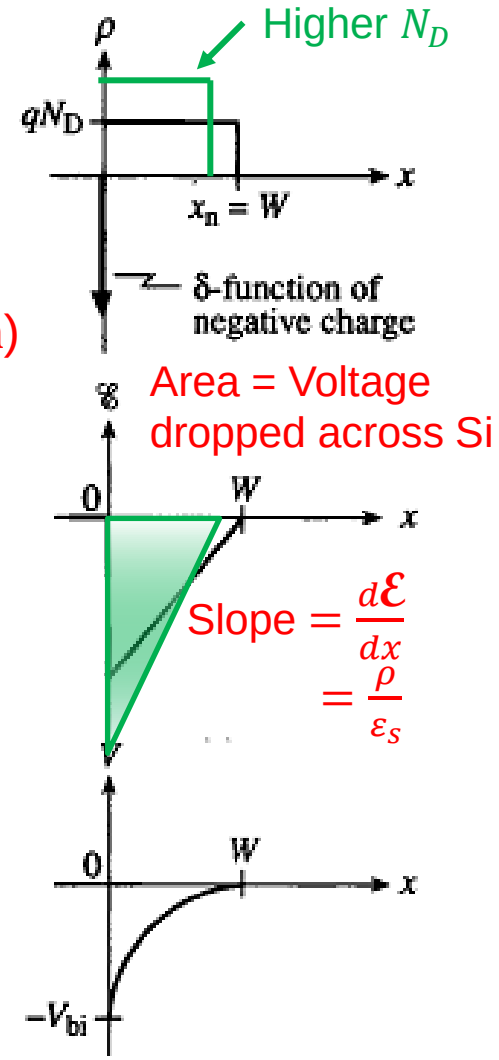
$$V(x) = \frac{-qN_D}{2K_S\varepsilon_0}(W - x)^2$$

11.7 for Si ($\varepsilon_{Si} = 10^{-12}$F/cm)

At $x = 0$, $V = -V_{bi}$

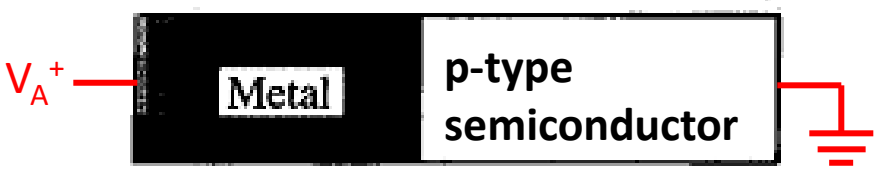
$$\Rightarrow \quad W = \sqrt{\frac{2\varepsilon_s V_{bi}}{qN_D}}$$

- ***W* decreases with increasing N_D**

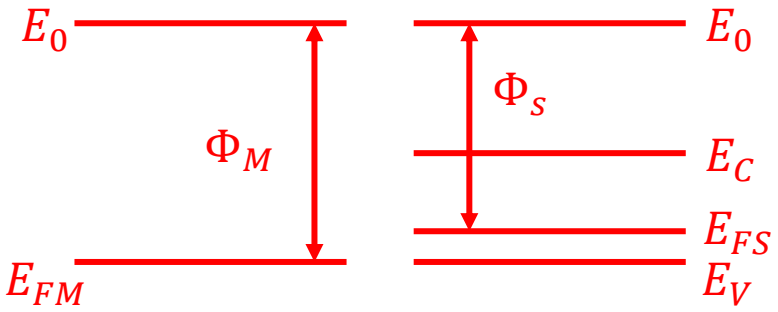


R.F. Pierret, *Semiconductor Fundamentals*, Fig. 14.4

Ideal M-S Contact: $\Phi_M > \Phi_S$, p-type

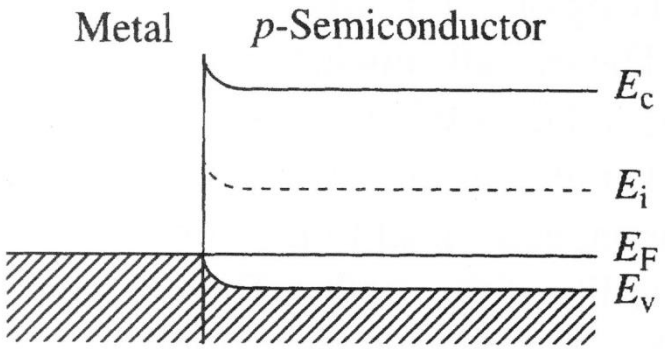


Band diagram instantly after contact formation:



Ohmic contact

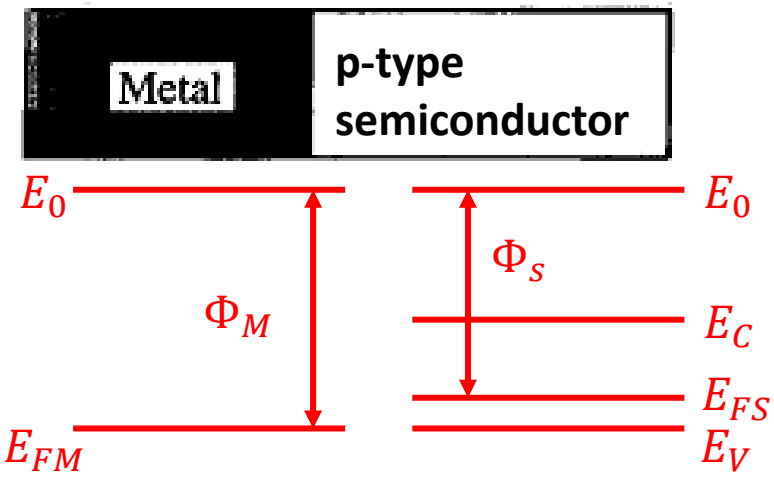
Equilibrium band diagram:



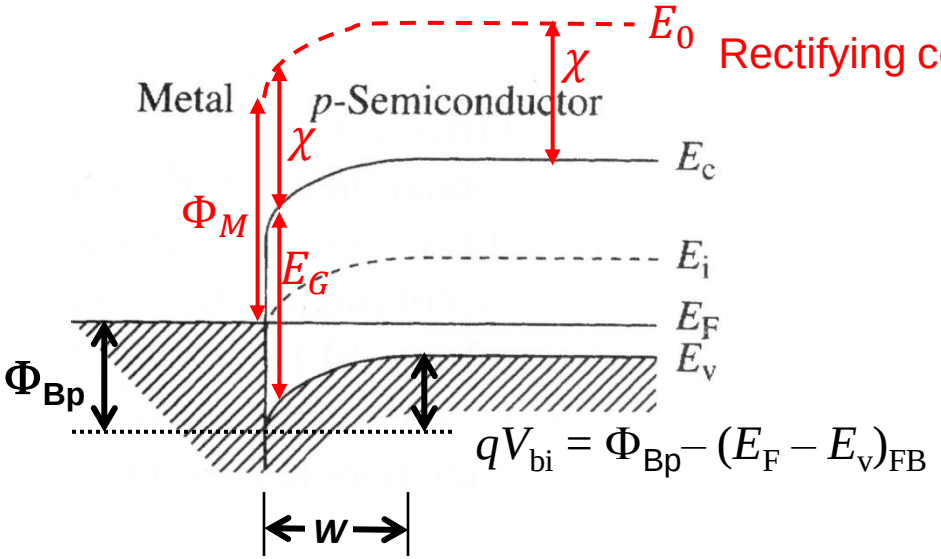
R.F. Pierret, *Semiconductor Fundamentals*, p. 482

Ideal M-S Contact: $\Phi_M < \Phi_S$, p-type

Band diagram instantly after contact formation:



Equilibrium band diagram:



Rectifying contact

***Schottky Barrier Height*:**

$$\Phi_{Bp} = \chi + E_G - \Phi_M$$

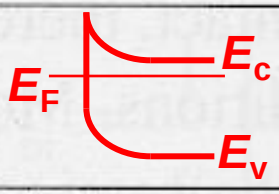
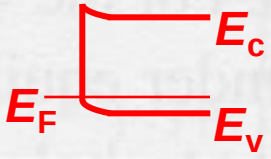
Note: $\Phi_{BP} + \Phi_{BN} = E_G$

$\sim \frac{1}{3} E_G \quad \sim \frac{2}{3} E_G$

R.F. Pierret, *Semiconductor Fundamentals*, p. 482

Summary

Table 14.1 Electrical Nature of Ideal MS Contacts.

	n-type Semiconductor	*p-type Semiconductor*
$\Phi_M > \Phi_S$	 Rectifying	 Ohmic
$\Phi_M < \Phi_S$	 Ohmic	 Rectifying

R.F. Pierret, *Semiconductor Fundamentals*, p. 481

Summary: Rectifying Contacts

- **Schottky barrier height, Φ_B:**
  - Energy barrier that must be surmounted in order for a carrier in the metal to enter the semiconductor
- **Built-in potential, qV_{bi}:**

$\Phi_{Bn}-(E_C-E_F)_{FB}$ for n-type Si, $\Phi_{Bp}-(E_F-E_v)_{FB}$ for p-type Si

  - Ideally qV_{bi} is equal to the work function difference between the metal and semiconductor.

    In practice, for Si:

$$\Phi_{Bn} \cong (2/3)E_G \text{ and } \Phi_{Bp} \cong (1/3)E_G$$

Acknowledgement

- Modern Semiconductor Devices for Integrated Circuits by C. Hu (Prentice Hall, 2009)
- Semiconductor Device Fundamentals by R. F. Pierret (Addison Wesley, 1996)
- Professor Tsu-Jae King Liu EE130/230A notes